

Update on 2025 post-TS tracking efficiencies

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Heidelberg University

16th June 2026



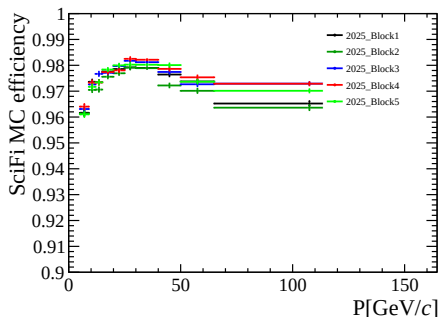
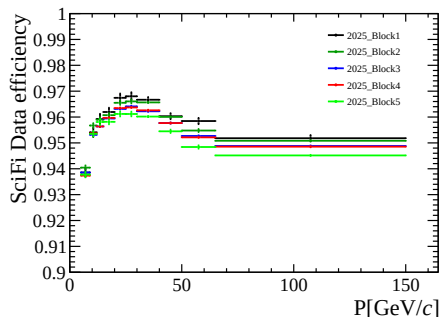
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HEIDELBERG**
ZUKUNFT
SEIT 1386

Progress update:

- ① Tracking efficiency tables presented in RTA WP4 two weeks ago
 - No objections or issues raised
 - Tables pushed to GitLab repo for analysts to use on 4th June
- ② Today: 2025 post-TS tables
 - MC request with priority 1a finished very quickly
 - Most fits converged without issues
 - I would like to release these tables as well
- ③ Plan to present a quick update in RTA WP4 on Thursday

2025 tracking efficiencies

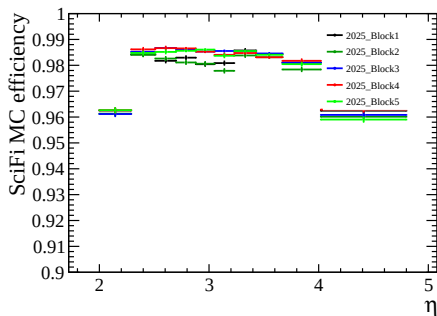
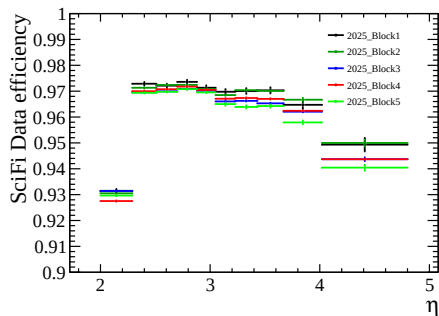
Comparison of tracking efficiencies in 2025 data blocks



SciFi efficiencies dropping slowly over time in data

2025 tracking efficiencies

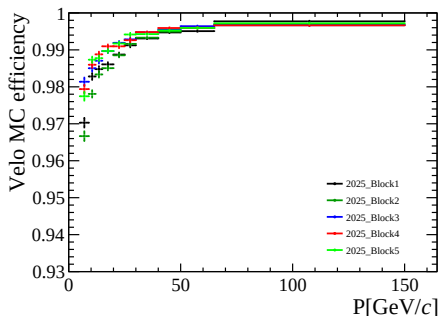
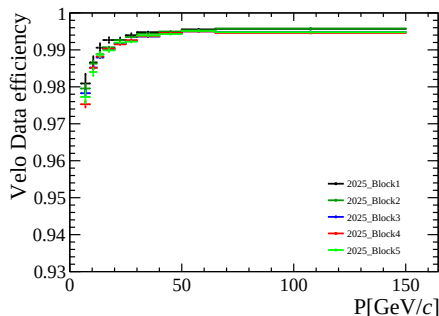
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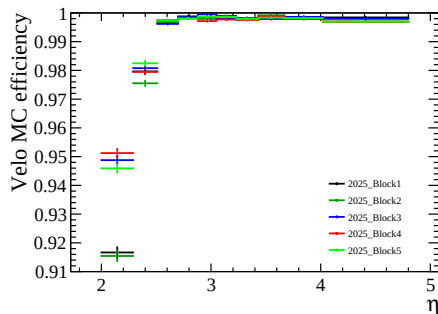
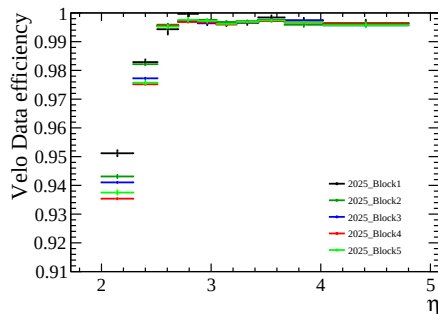
Comparison of tracking efficiencies in 2025 data blocks



Velo efficiencies underestimated in blocks 1/2, but not blocks 3/4/5

2025 tracking efficiencies

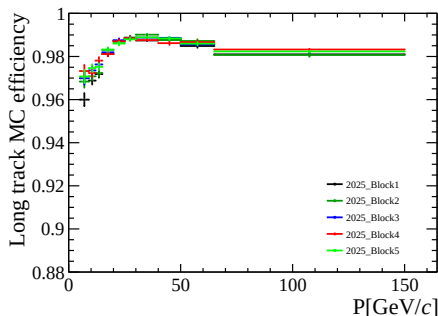
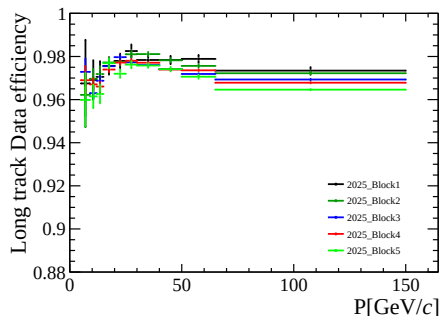
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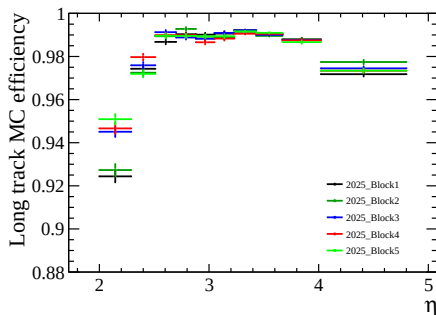
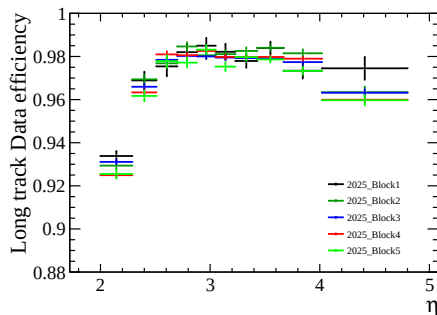
Comparison of tracking efficiencies in 2025 data blocks



MuonUT efficiencies look consistent, but lower statistics

2025 tracking efficiencies

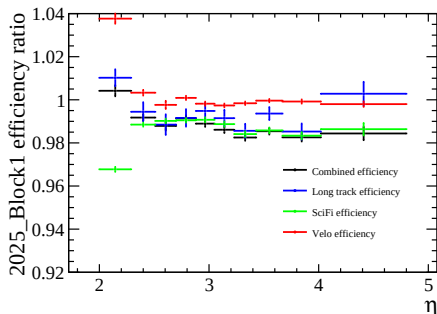
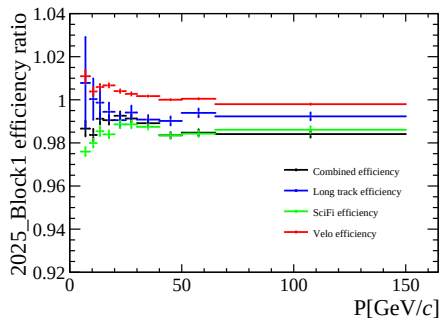
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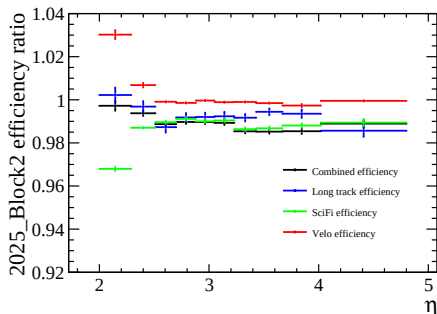
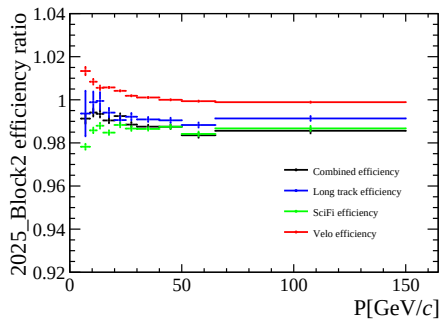
2025 tracking efficiencies

Data-MC ratio in 2025 block 1



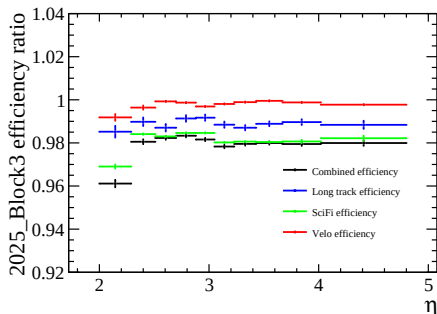
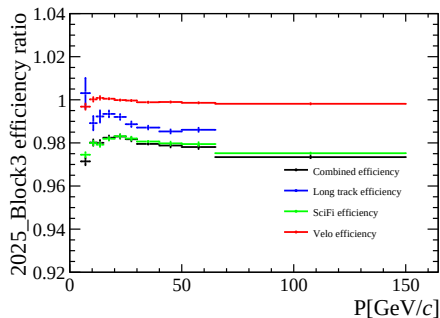
2025 tracking efficiencies

Data-MC ratio in 2025 block 2



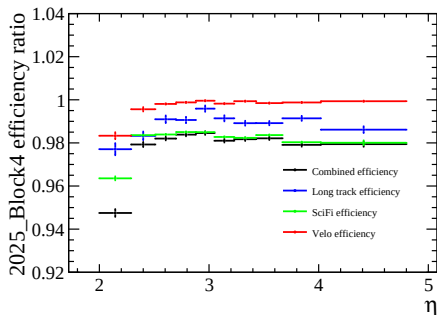
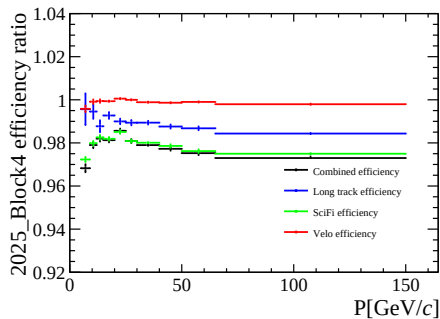
2025 tracking efficiencies

Data-MC ratio in 2025 block 3



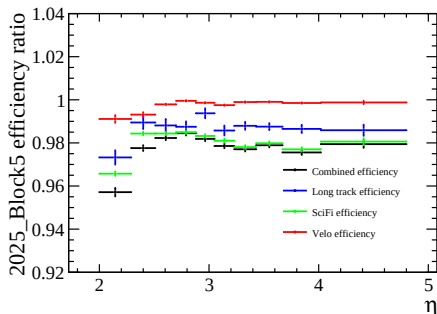
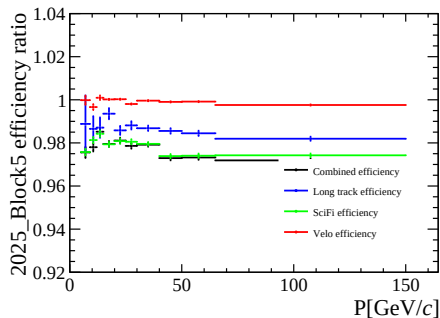
2025 tracking efficiencies

Data-MC ratio in 2025 block 4

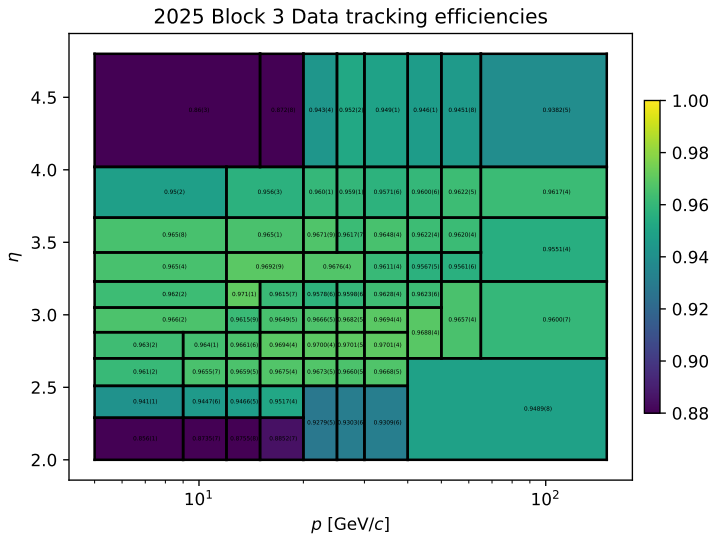


2025 tracking efficiencies

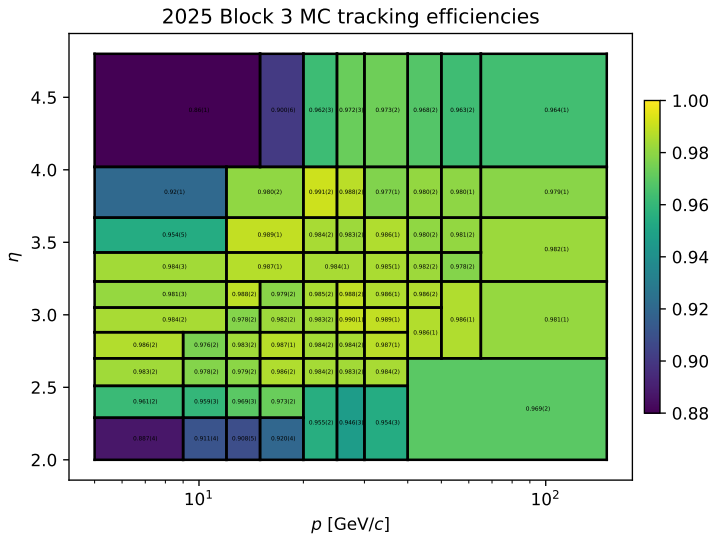
Data-MC ratio in 2025 block 5



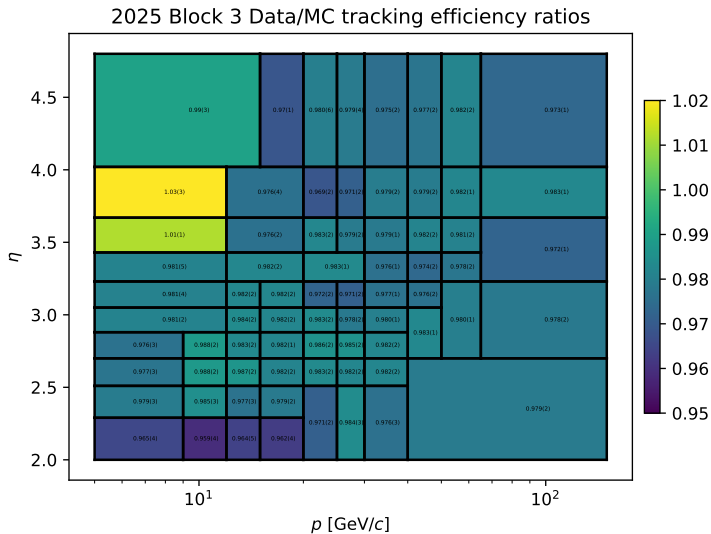
2025 pre-TS tracking efficiency corrections



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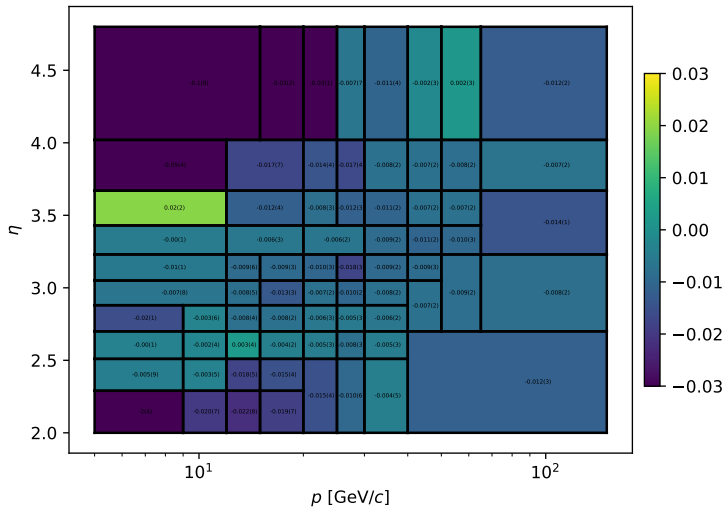


2025 pre-TS tracking efficiency corrections

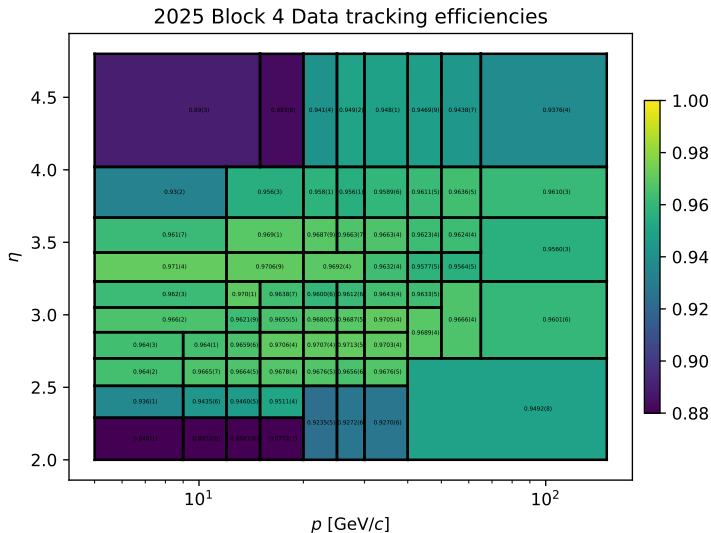


2025 pre-TS tracking efficiency corrections

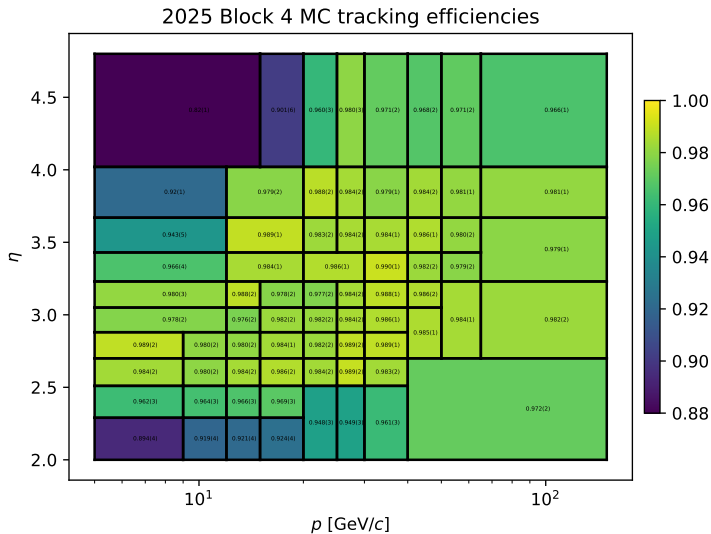
2025 Block 3 Data/MC Combined vs MuonUT ratio difference



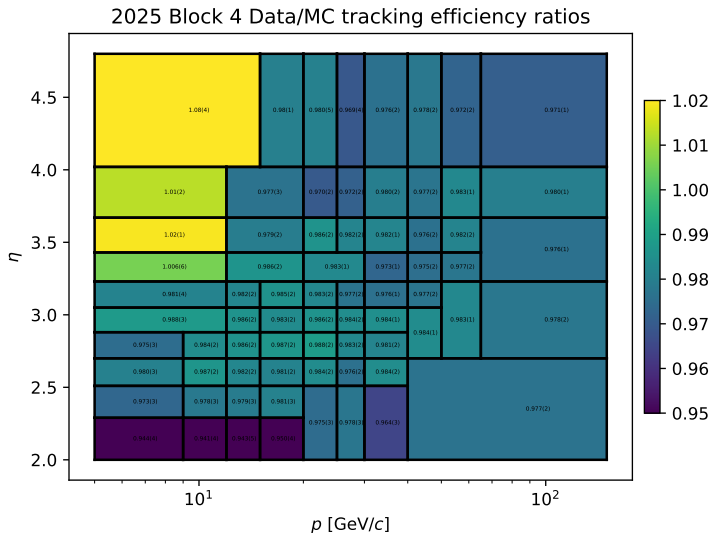
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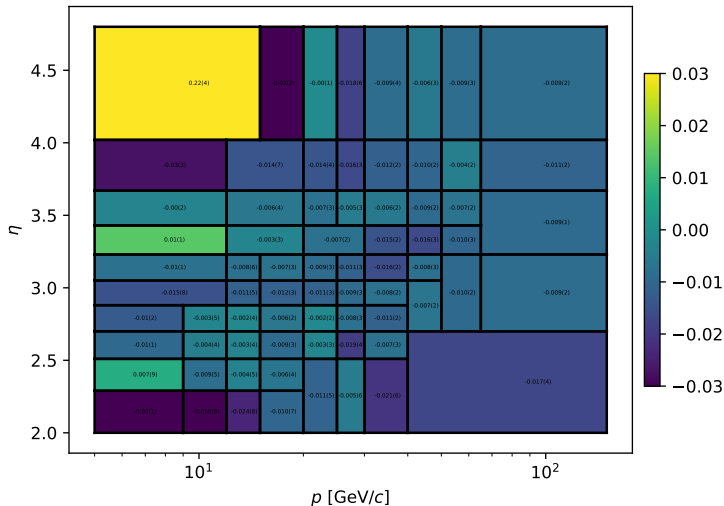


2025 pre-TS tracking efficiency corrections



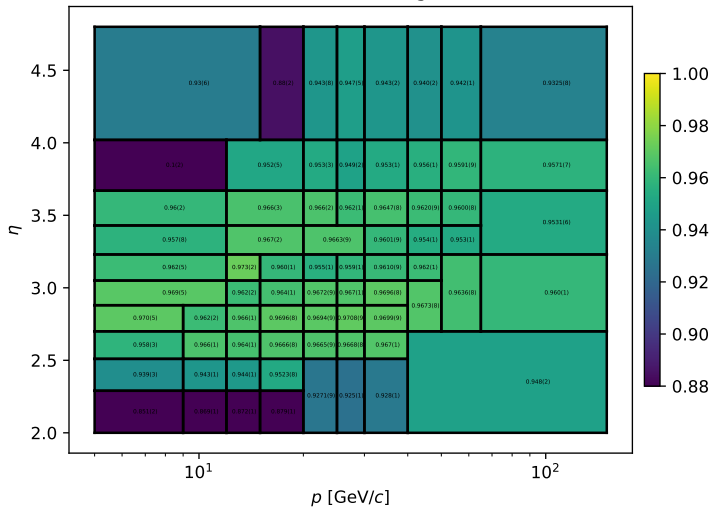
2025 pre-TS tracking efficiency corrections

2025 Block 4 Data/MC Combined vs MuonUT ratio difference

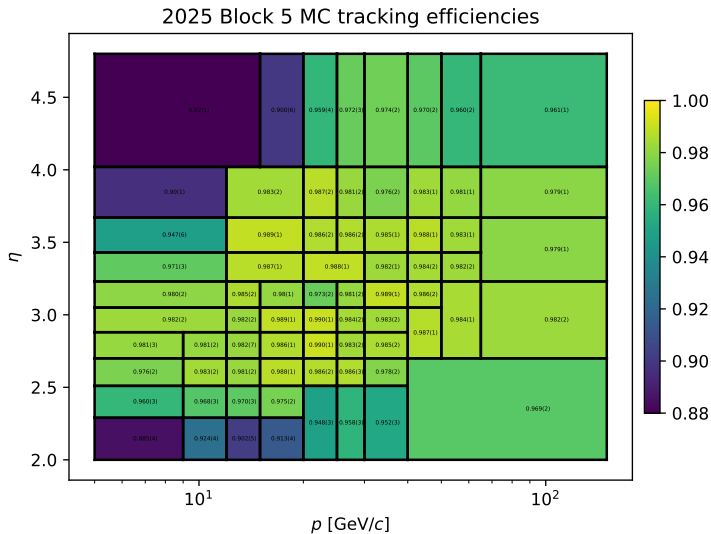


2025 pre-TS tracking efficiency corrections

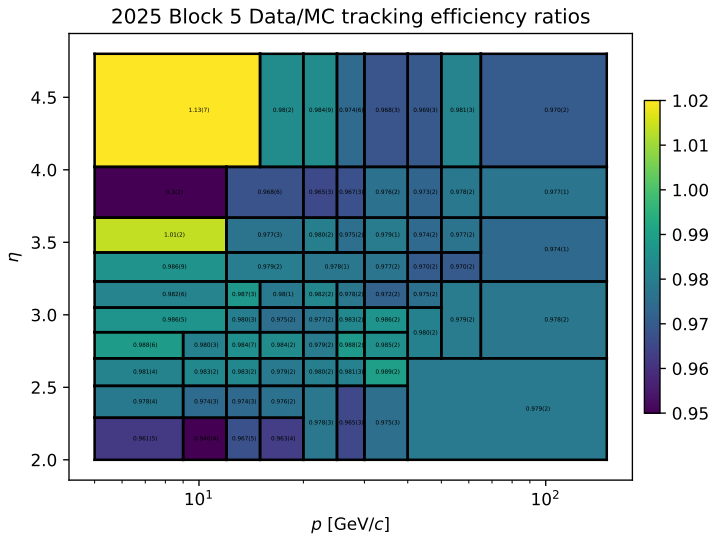
2025 Block 5 Data tracking efficiencies



2025 pre-TS tracking efficiency corrections

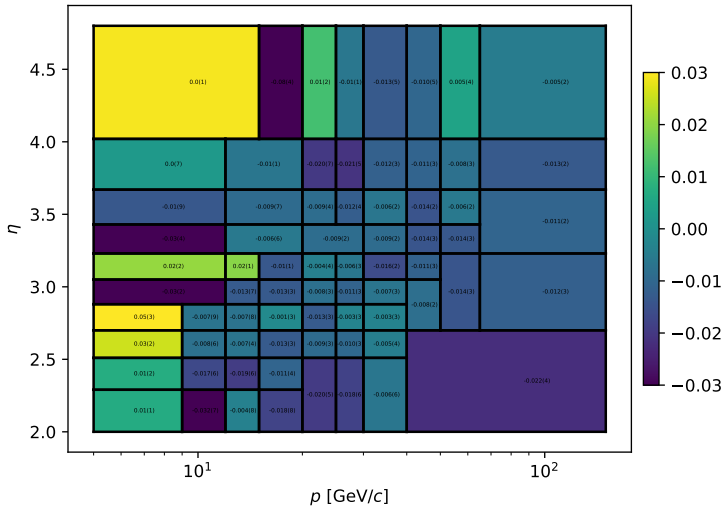


2025 pre-TS tracking efficiency corrections



2025 pre-TS tracking efficiency corrections

2025 Block 5 Data/MC Combined vs MuonUT ratio difference



Next steps:

- ① Large 2024 production is done
 - Produce v1 tables with $\mu^{\pm}$, μ^{+} , μ^{-} using new binning, look into reweighting multiplicity
 - Will probably take a while, as I'm quite busy with analysis as well
- ② Look into block 1b MC